BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 4481 / 4482

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q1. Q. 1

Q1. Classification of elements in the modern periodic table is based on

(A) (B)

(C) (D)

Q2. Chile saltpetre has the chemical formula

(A) (B)

(C) (D)

Q3. Aluminium oxide is

(A) (B)

(C) (D)

Q4. Aqua Regia can dissolve noble metals due to the formation of

(A) (B)

(C) (D)

Q5. Which halogen occurs naturally in a positive oxidation state

(A) (B)

(C) (D)

Q6. Group VI B of Transition elements contain

(A) (B)

(C) (D)

Q7. Select from the following the one which is alcohol

(A) (B)

(C) (D)

Q8. Preparation of vegetable ghee involves

(A) (B)

(C) (D)

Q9. Which compound is the most reactive one

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp fully covers Q9 option (C); marked [illegible in source scan].

Q10. Nucleophile is usually

(A) (B)

(C) (D)

Q11. Which of the following compound is called Universal Solvant?

(A) (B)

(C) (D)

Q12. Acetone react with HCN to form cyanohydrin. It is an example of

(A) (B)

(C) (D)

Q13. Which acid is used in manufacture of synthetic fiber

(A) (B)

(C) (D)

Q14. Which one of the following element is not present in all proteins.

(A) (B)

(C) (D)

Q15. Micronutrient element are required in Quantity

(A) (B)

(C) (D)

Q16. The pH range of acid rain is

(A) (B)

(C) (D)

Q17. Which one of the following substance cause acid rain?

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) Why 2nd ionization energy is greater than 1st ionization energy?

(ii) Why is diamond non-conductor but graphite is conductor?

(iii) What is general trend for the solubility of sulphates of alkaline earth metals?

(iv) Why lime is added to acidic soil?

(v) How does $K_2Cr_2O_7$ oxidize the H_2S and $FeSO_4$?

(vi) How zinc coating prevents iron from corrosion?

(vii) How antiknocking agents are prepared from alkyl halides?

(viii) What is wurtz synthesis?

(ix) What is function of nucleic acid?

(x) How temperature affects the activity of enzymes?

(xi) How triglycerides are hydrolyzed?

(xii) What is meant by dry cleaning in paper manufacturing?

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Why N_2O is called laughing gas?

(ii) Write down any four uses of Nitric acid.

(iii) What are Freons and Teflon?

(iv) Why HF is weaker acid than HCl? Justify.

(v) Define aromatic compound by giving two examples.

(vi) How vital theory was rejected?

(vii) Convert methane into methyl alcohol.

(viii) Write down the industrial preparation of Ethyne.

(ix) Mention the four physical properties of Ethene.

(x) Mention any two conditions which are required for the formation of smog.

(xi) Write short note on chemical oxygen demand. (COD).

(xii) Is detergent are threat to aquatic life? Justify.

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) Why are liquid silicones preferred over ordinary organic lubricants?

(ii) What is the action of an aqueous solution of borax on litmus?

(iii) How will you convert boric acid into borax and vice versa.

(iv) What is Wurtz-Fittig reaction?

(v) How bakelite is produced? Give reaction.

(vi) How methanol and ethanol can be differentiated?

(vii) What is Tollen's test? Give reaction.

(viii) How carboxylic acid is prepared from alkene?

(ix) Differentiate between essential and non-essential amino acids.

Q. Section ---------- II

Note: Attempt any three questions. (8 × 3 = 24)

Q5. 5. (a) What are the improvements made in the Mendeleev's periodic table?

(b) Describe the role of Gypsum in agriculture and industry. (Any four points of each)

Q6. 6. (a) What happens when bleaching powder reacts with (i) dil H_2SO_4 (ii) excess of conc. H_2SO_4 (iii) NH_3 (iv) CO_2

(b) What is meant by setting of cement? What are the reactions taking place in first twenty four hours and between one to seven days?

Q7. 7. (a) Discuss sp^2 – hybridization with a suitable example.

(b) How would you prepare following compounds from Grignard reagent? (i) 1-butanol (ii) 2-butanol (iii) Cyanogen chloride (iv) ethane

Q8. 8. (a) How is ethene prepared by Kolbe's electrolytic method. Give its mechanism also.

(b) Give the reactions of acetone with. (i) HCN (ii) NH_2OH (iii) NH_2NH_2 (iv) $NaHSO_3$

Q9. 9. (a) Describe oxidation reactions of Benzene and alkyl benzene.

(b) Explain the following terms (i) Absolute alcohol (ii) Methylated spirit (iii) Rectified spirit (iv) Denaturing of alcohols

GROUP – II

Q1. Q. 1

Q1. Which of the following metal does not form ionic hydride.

(A) (B)

(C) (D)

Q2. The mineral $CaSO_4.2H_2O$ has the general name.

(A) (B)

(C) (D)

Q3. Which of the following element is not present abundantly in earth's crust?

(A) (B)

(C) (D)

Q4. Which catalyst is used in contact process

(A) (B)

(C) (D)

Q5. Which halogen will react spontaneously with Au to produce Au^{+3}?

(A) (B)

(C) (D)

Q6. The strength of binding energy of transition elements depends on

(A) (B)

(C) (D)

Q7. In t-butyl alcohol, the tertiary carbon is bonded to

(A) (B)

(C) (D)

Q8. Synthetic rubber is made by polymerization of

(A) (B)

(C) (D)

Q9. The electrophile in aromatic sulphonation is

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp fully covers Q9 option (C); marked [illegible in source scan].

Q10. Ethyl magnesium bromide react with water to form

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp partially overlaps Q10; the question and options are still fully readable.

Q11. Rectified spirit contains alcohol about

(A) (B)

(C) (D)

Q12. The carbon atom of a carbonyl group is

(A) (B)

(C) (D)

Q13. Which of the following is not a fatty acid?

(A) (B)

(C) (D)

Q14. The reaction between fat and NaOH is called

(A) (B)

(C) (D)

Q15. Ammonium Nitrate fertilizer is not used for which crop.

(A) (B)

(C) (D)

Q16. Ecosystem is a smaller unit of

(A) (B)

(C) (D)

Q17. Which gas is not a pollutant

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) What improvements were made in Mendleev's periodic table?

(ii) The radius of Na is 157 pm while that of Na^+ is 95 pm. Why?

(iii) What happens when Lithium carbonate and Lithium nitrates are heated?

(iv) Why is calcium important for plant growth?

(v) How does acidified $KMnO_4$ Oxidize $FeSO_4$ and oxalic acid?

(vi) Why do the transition element show variable valency?

(vii) Why alkyl Iodides are the most reactive among alkyl halides?

(viii) What is leaving group. Give two examples.

(ix) Write name of factors with affect the enzyme activity.

(x) What are the Differences between DNA and RNA?

(xi) Define acid number and Iodine number with example.

(xii) Why wet process is favourable for manufacturing of cement in Pakistan?

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) NO_2 is a strong oxidizing agent. Prove the truth of this statement giving example.

(ii) Write down any four uses of sulphuric acid.

(iii) Why HF is weaker acid than HCL? Justify.

(iv) What are Freons and Teflon?

(v) Draw the various isomer of Pentane (C_5H_{12})

(vi) Why there is no free rotation around a double bond and a free rotation around a single bond. Justify.

(vii) How does ethyne react with (a) Halogen acid (b) Ammonical cuprous chloride

(viii) Write short note on acidity of Ethyne.

(ix) Write down any four uses of ethene.

(x) Mention any two conditions which are required for the formation of smog.

(xi) Is detergents are threat to aquatic life? Justify.

(xii) Write short note on dissolved oxygen (DO).

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) What is the alkanoic acid. How it is prepared from alkanol?

(ii) Justify the acidic and basic character of amino acids.

(iii) Prepare borax from (a) Colemanite (b) Boric acid

(iv) How "Al" reacts with hydrogen and halogen?

(v) Justify "CO_2" is non polar in nature.

(vi) Write the name and structure of two aromatic compounds containing two benzene rings.

(vii) Why phenol is more acidic than alcohol?

(viii) Ethanol has highest boiling point than di ethyl ether.

(ix) Write the oxidation reaction of aldehyde with (a) $K_2Cr_2O_7$ / H_2SO_4 (b) Tollen's reagent

Q. Section ---------- II

Note: Attempt any three questions. (8 × 3 = 24)

Q5. 5. (a) Define periodic table. What improvements are made in Mendeleev's periodic table?

(b) How sodium metal is produced by Down's cell. Sketch labelled Down's cell.

Q6. 6. (a) What are "Disproportionation reactions"? Explain your answer with the reaction of chlorine with hot and cold NaOH.

(b) What do you understand by the term "Setting of cement"? Also discuss the reactions taking place in first 24 hours?

Q7. 7. (a) What are the various sources of organic compounds? Give significance of Coal amongst these sources.

(b) Using ethyl bromide as a starting material how would you prepare the following compounds? (i) n-butane (ii) Ethyl alcohol (iii) Ethene (iv) Ethane

Q8. 8. (a) Explain the following reactions. (i) Ozonolysis of ethene (ii) Oxidation of ethyne by $KMnO_4$

(b) What is Cannizzaro's reaction? Write down its mechanism.

Q9. 9. (a) Describe the stability of Benzene molecule by estimating Heats of Hydrogenation.

(b) Explain the following terms using ethyl alcohol as an example. (i) Oxidation (ii) Dehydration (iii) Esterification (iv) Ether formation